Report

Project Title:
Movie Recommendation System

Dataset : 1Million Data

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1. Introduction

This project is a hybrid movie recommendation system designed to generate personalized movie suggestions using multiple machine learning approaches. It combines collaborative filtering, neural networks, and clustering techniques to improve recommendation quality.

2. Objective

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3. Method Used

3.1 Collaborative Filtering (ALS)

Matrix factorization technique used to learn latent features of users and movies.

3.2 Neural Collaborative Filtering (NCF)

Deep learning model that captures non-linear relationships between users and items.

3.3 Clustering Approach

Groups similar users/items to improve recommendation relevance.

3.4 Hybrid Model

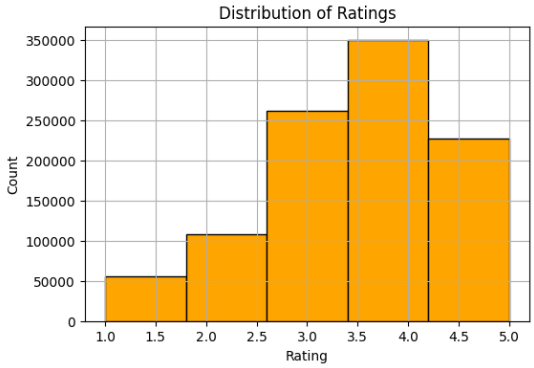
Combines ALS + NCF + Clustering to produce final recommendations.

4. Exploratory Data Analysis (EDA)

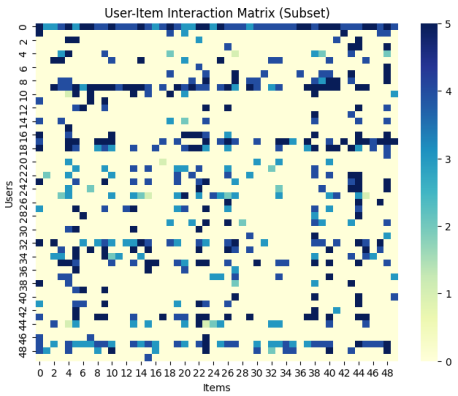
Key insights from the dataset:

- Rating distribution is centered around mid-high values
- Some users are highly active compared to others
- A small number of movies receive most of the ratings

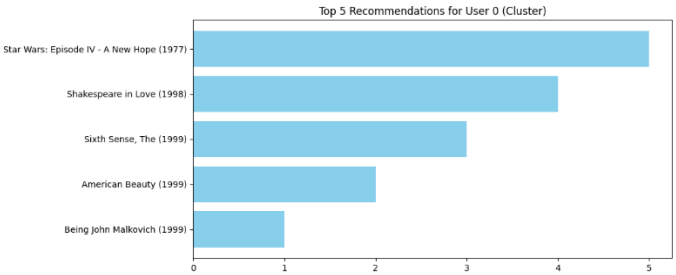
Rating Distrbution

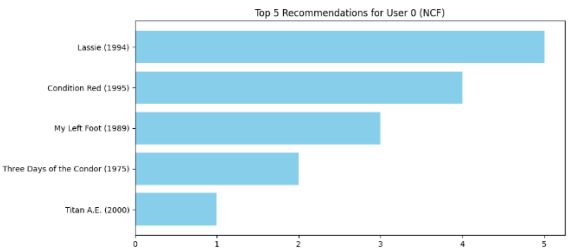


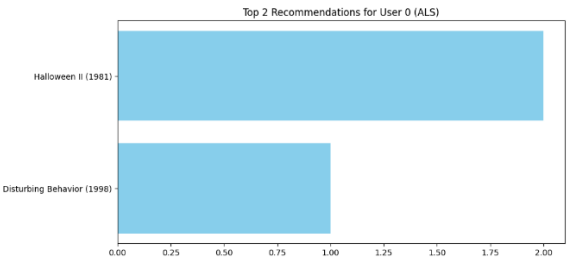
User activity



Movie Popularity







5. Results

The hybrid model improves recommendation quality by combining strengths of different approaches:

- ALS: Strong for latent factor learning
- NCF: Captures complex patterns
- Clustering: Improves grouping similarity

Final Outcome:

The hybrid system provides more stable and accurate recommendations compared to single models.

6. Tech Stack

Python

Pandas / NumPy

Scikit-learn

TensorFlow / PyTorch

Matplotlib / Seaborn

7. Conclusion

This project demonstrates how combining multiple recommendation techniques can significantly improve prediction quality and user personalization. It serves as a foundation for real-world recommendation systems like Netflix or Spotify.